Supplementary Material for "Cast-Break-Free Demand Response of Electric Arc Furnace Fleets: Exact Counting Polytopes and Certified Rolling Dispatch"

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This document proves Theorems 1 to 4 and gives the boundary rows of (3) and the example for Theorem 1(ii). Equations without the prefix S are numbered as in the paper.

I. Boundary Rows of the Counting Polytope

Heats are indexed by start from the last starts of the live state, so heats $1,\dots,n$ are the last heats of the furnaces idle at the notice and casts $1,\dots,N^{\mathrm{C}}$ are the last casts of the casters. Under first-in first-out casting, the Q waiting ladles feed casts $N^{\mathrm{C}}+1,\dots,N^{\mathrm{C}}+Q$, and the ladle of heat $k>n$ feeds cast $k+q$, with $q=Q+N^{\mathrm{C}}-n$. The counters before $t=0$ are fixed, (3d) is imposed for $t\ge 0$, (3e) for $x(t)>n$, and

$$y(\tau_j+w)\ge N^{\mathrm{C}}+j,\qquad j=1,\dots,Q, \tag{S1}$$

where $\tau_j\le 0$ is the tap of the j-th oldest waiting ladle. Beyond H, (3b) and (3c) use $x(\min\{\cdot,H\})$ and $y(\min\{\cdot,H\})$, and the other rows are dropped.

II. Proof of Theorem 1

For integer counters, let $s_k=\min\{t: x(t)\ge k\}$, $z_k=\min\{t: y(t)\ge k\}$ and $v_k=s_k+d(s_k)$. Then $x(t)\ge k\Leftrightarrow s_k\le t$, $y(t)\ge k\Leftrightarrow z_k\le t$, and, as $t+d(t)$ is non-decreasing, $x(\theta(t))\ge k\Leftrightarrow v_k\le t$.

Lemma S1. *For integer counters satisfying* (3a)*, rows* (3b) *to* (3f) *hold if and only if, for all starts in the horizon,*

$$s_{k+N^{\mathrm{E}}}\ge v_k+g, \tag{S2a}$$
$$z_k+c(z_k)\le z_{k+N^{\mathrm{C}}}\le z_k+c(z_k)+b, \tag{S2b}$$
$$v_k\le z_{k+q}\le v_k+w,\quad k>n. \tag{S2c}$$

Proof. Rows (3b), (3c), (3f), (3d) and (3e) at $t=s_k$, z_k, z_k, z_{k+q} and s_k give

$$x(v_k+g-1)\le k-1+N^{\mathrm{E}},\ y\big(z_k+c(z_k)-1\big)\le k-1+N^{\mathrm{C}},$$
$$y\big(z_k+c(z_k)+b\big)\ge k+N^{\mathrm{C}},\quad x\big(\theta(z_{k+q})\big)\ge k,$$
$$y(v_k+w)\ge k+q,$$

which are (S2a) to (S2c). Conversely, for (3b) at t, let $k=x(t-1)+1$. Then $s_k\ge t$, and (S2a) gives $s_{k+N^{\mathrm{E}}}\ge t+d(t)+g$, that is, $x(t+d(t)+g-1)\le x(t-1)+N^{\mathrm{E}}$. Rows (3c), (3f), (3d) and (3e) follow in the same way with $k=y(t-1)+1$, $y(t)$, $y(t)-q$ and $x(t)$, and (3d) holds trivially if $y(t)\le N^{\mathrm{C}}+Q$, since $x(\theta(t))\ge n$ for $t\ge 0$. □

Proof of Theorem 1. (i) Let a schedule be zero-break. A furnace that starts a heat at $s\in[t,\,t+d(t)+g-1]$ starts its next heat at $s'\ge s+d(s)+g\ge t+d(t)+g$ by (1d), so each furnace starts at most one heat in this window, which gives (3b), and (1e) gives (3c) and (3f) in the same way. The $y(t)-N^{\mathrm{C}}$ casts started by t after the last casts need distinct ladles, waiting or tapped in $[0,t]$, and by (1f) the waiting ladles and those of the heats $k>n$ started by t are cast by $t+d(t)+w$. Hence

$$y(t)-N^{\mathrm{C}}\le Q+x\big(\theta(t)\big)-n,$$
$$y\big(t+d(t)+w\big)-N^{\mathrm{C}}\ge Q+x(t)-n,$$

which are (3d) and (3e), and (S1) holds likewise. As the counters do not decrease after H, the capped rows also hold. Conversely, the starts of an integer point of $\mathcal{F}$ satisfy (S2) by Lemma S1. Under the round-robin assignment, heat $k+N^{\mathrm{E}}$ is the next heat on the furnace of heat k, cast $k+N^{\mathrm{C}}$ is the next cast on the caster of cast k, and cast $k+q$ receives the ladle of heat k, so (S2) and (S1) are (1d) to (1f).

(ii) Each row of $\mathcal{F}$ is a difference constraint with an integer right side, so the constraint matrix is totally unimodular [1], and $\mathcal{F}$, bounded by (3b) and (3c), is an integral polytope. As (1c) is linear, it maps $\mathcal{F}=\mathrm{conv}(\mathcal{F}\cap\mathbb{Z})$ onto the convex hull of the block loads of the zero-break schedules. □

Example S1. The image in Theorem 1(ii) need not be a g-polymatroid. Let $N^{\mathrm{E}}=N^{\mathrm{C}}=1$, $d=2$, $g=0$, $c=3$ and $b=w=1$, with an idle furnace, no waiting ladle, a cast in progress that ends at $t=1$, $t_0=0$ and $H\ge 10$. Then $s_1=0$, $z_1=2$, $z_{k+1}-z_k\in\{3,4\}$, $z_k-v_k\in\{0,1\}$ and $s_{k+1}\ge v_k$. Enumerating these schedules gives, in units of $P/3$, the load vectors (ℓ_1,ℓ_2,ℓ_3) of the first three blocks

$$(2,2,1),\ (2,2,2),\ (2,2,3),\ (2,3,1),\ (2,3,2),$$
$$(3,1,2),\ (3,1,3),\ (3,2,1),\ (3,2,2).$$

The function $\ell_1+2\ell_2+\ell_3$ equals 7 at $(2,2,1)$, with $s=(0,3,8)$, and at $(3,1,2)$, with $s=(0,2,6,9)$, and is at least 8 elsewhere. These two vectors thus span an edge of the projected hull in direction $(1,-1,1)$, whereas every edge of a g-polymatroid

and of its coordinate projections is parallel to some e_j or $e_j - e_k$ [2].

III. Proof of Theorem 2

With $p_t = \underline{P} + (\overline{P} - \underline{P})\omega_t$, (4) becomes

$$\kappa(s, v) = \underline{P} \sum_{t=s}^{v-1} \pi_t + (\overline{P} - \underline{P})\, \phi\big(\{s, \dots, v-1\}, \nu_{v-s}\big), \quad \text{(S3)}$$

where $\phi(I, \nu) = \min\{\sum_{t\in I} \pi_t \omega_t : \omega \in [0,1]^I,\ \sum_{t\in I} \omega_t = \nu\}$ is convex in ν on $[0, |I|]$, $\nu_d = (E - d\underline{P})/(\overline{P} - \underline{P})$, and $\nu_{d+1} = \nu_d - \delta$ with $\delta = \underline{P}/(\overline{P} - \underline{P})$. The first term of (S3) is modular in (s, v) and cancels below.

Let $(s, v+1)$ and $(s+1, v)$ be admissible, $I = \{s+1, \dots, v-1\}$, $\nu = \nu_{v-s}$, and let α and α' be the weights of s and v in an optimal solution of $\phi(I \cup \{s, v\}, \nu - \delta)$. Then

$$\begin{aligned}
&\phi(I \cup \{s\}, \nu) + \phi(I \cup \{v\}, \nu) \\
&\quad \le \pi_s \alpha + \pi_v \alpha' + \phi(I, \nu - \alpha) + \phi(I, \nu - \alpha') \\
&\quad \le \pi_s \alpha + \pi_v \alpha' + \phi(I, \nu - \delta - \alpha - \alpha') + \phi(I, \nu + \delta) \\
&\quad = \phi(I \cup \{s, v\}, \nu - \delta) + \phi(I, \nu + \delta),
\end{aligned}$$

where the second step is convexity, as $\nu - \delta - \alpha - \alpha' \le \nu - \alpha, \nu - \alpha' \le \nu + \delta$ have equal sums and lie in $[0, |I|]$ because $\alpha + \alpha' \le \nu - \delta$ and $(s+1, v)$ is admissible. By (S3),

$$\kappa(s, v) + \kappa(s+1, v+1) \le \kappa(s, v+1) + \kappa(s+1, v). \quad \text{(S4)}$$

Admissible durations form an interval, so summing (S4) over $s_1 \le s < s_2$ and $v_1 \le v < v_2$ gives the Monge inequality $\kappa(s_1, v_1) + \kappa(s_2, v_2) \le \kappa(s_1, v_2) + \kappa(s_2, v_1)$ for all admissible nested arcs (s_1, v_2) and (s_2, v_1).

Heats in progress are fixed and only shift the integer right sides of the rows. Moving $\varepsilon = \min\{\xi(s_1, v_2), \xi(s_2, v_1)\}$ from a nested pair to (s_1, v_1) and (s_2, v_2) keeps x, $\hat{x}$ and every row, does not raise the cost, and lowers $\sum \xi(s, v)(v - s)^2$ by $2\varepsilon(s_2 - s_1)(v_2 - v_1)$. Some optimal ξ thus has no nested pair, and its support orders as $a_1, \dots, a_R$ with non-decreasing starts and taps. For $u \in [0, 1)$ and every row $a - a' \le h$ with $h \in \mathbb{Z}$,

$$\lfloor a + u \rfloor - \lfloor a' + u \rfloor \le \lceil a - a' \rceil \le h, \quad \text{(S5)}$$

so the shifted floors of x, $\hat{x}$ and y keep every row [3]. With $M_j = \sum_{i \le j} \xi(a_i)$, the k-th rounded start and tap both lie on a_{j_k}, $j_k = \min\{j : M_j \ge k - u\}$, so the rounding is a zero-break schedule on arcs of the support. For u uniform on $[0, 1)$, arc a_j carries $\xi(a_j)$ heats on average and $\mathbb{E}\lfloor y + u \rfloor = y$, so some u gives an integral schedule that costs at most the LP value, which is therefore optimal. □

IV. Proof of Theorem 3

1) Part (i): By (S5), the shifted floors (8) of a point of $\mathcal{F}_i$ lie in $\mathcal{F}_i \cap \mathbb{Z}$ and are zero-break schedules by Theorem 1. For u uniform on $[0, 1)$, $\mathbb{E}\lfloor a + u \rfloor = a$, and each running count rounds to $\lfloor r_i(t) \rfloor$ or $\lceil r_i(t) \rceil$, so $\mathbb{E}\,\ell_{i,\beta} = \ell^{\mathrm{LP}}_{i,\beta}$ and $\ell_{i,\beta}$ ranges over an interval of length P_i. Markov's inequality, the independence of the shifts and the union bound give

$$\Pr\left\{ \max_{\beta} \left(\sum_i \ell_{i,\beta} - \bar{L}_\beta \right) \ge C^{\mathrm{LP}} + \Delta \right\} \le U(\Delta, \lambda).$$

Let U_j be U with the shifts of the first j plants of $\mathcal{S}$ fixed. As U_j averages U_{j+1} over the next shift, a minimizing shift gives $U_{j+1} \le U_j$, and $U_{|\mathcal{S}|} = \sum_\beta \exp\big(\lambda(\sum_i \ell_{i,\beta} - \bar{L}_\beta - C^{\mathrm{LP}} - \Delta)\big) < 1$ puts every block below $C^{\mathrm{LP}} + \Delta$. The floors of plant i change only where u_i crosses one minus the fractional part of one of the $O(m)$ counters in its block loads, so every expectation in (9) has $O(m)$ terms, and the rounding is polynomial.

At an optimal vertex, let G_i be the minimal face of $\mathcal{F}_i$ containing (x_i, y_i). The vertex is the only point of $G = \prod_i G_i \times \mathbb{R}$ near it at which the tight rows of (1b) are equalities, so $1 + \sum_i \dim G_i = \dim G \le m$. As $\mathcal{F}_i$ is integral, $\dim G_i \ge 1$ for $i \in \mathcal{S}$, and $|\mathcal{S}| \le m - 1$ [4]. By Hoeffding's lemma, each expectation in (9) is at most $e^{\lambda^2 P_i^2/8}$, so with $C^{\mathrm{LP}}_\beta \le C^{\mathrm{LP}}$ and $\lambda = 4\Delta/\sum_{i\in\mathcal{S}} P_i^2$,

$$U(\Delta, \lambda) \le m\, e^{-2\Delta^2/\sum_{i\in\mathcal{S}} P_i^2} \le m\, e^{-2\Delta^2/((m-1)P^2_{\max})}, \quad \text{(S6)}$$

which is below one for $\Delta > \big(\frac{1}{2}(m-1)\ln m\big)^{1/2} P_{\max}$.

2) Part (ii): Given sets $T_1, \dots, T_M \subseteq \{1, \dots, N\}$, give T_k the blocks A_k of slots $6k$ to $6k+2$ and B_k of slots $6k+3$ to $6k+5$, so $m = 2M$, and give element i a plant with $N^{\mathrm{E}} = N^{\mathrm{C}} = 1$, power $P = P_{\max}$, $g = w = 0$, $b = 1$ and $d(s) = 3, 4, 7, 6, 5, 4$ for $s \equiv 0, \dots, 5 \pmod 6$. Then $s + d(s)$ is non-decreasing and taps fall on slots $\equiv 3$ or $5 \pmod 6$. A cast started at $6k + 3$ or $6k + 5$ with $i \in T_k$ lasts until the same offset in the next set containing i, and a lead-in before the window lets plant i choose the offset of its first cast.

As $w = 0$ and $b = 1$, each cast starts at the end of the previous one, so plant i keeps one offset, $\chi_i = +1$ for 3 and $\chi_i = -1$ for 5. A tap at $6k+3$ needs a heat covering A_k, and a tap at $6k+5$ the heat in slots $6k+1$ to $6k+4$, so every schedule of plant i loads each block at least as much as the one that adds $P(5 + \chi_i)/6$ to A_k and $P(1 - \chi_i)/3$ to B_k for each k with $i \in T_k$. With $\bar{L}_{A_k} = 5P|T_k|/6 - C_0$, $\bar{L}_{B_k} = P|T_k|/3 - C_0$ and $\sigma_k = \sum_{i\in T_k} \chi_i$, the excesses are $C_0 + P\sigma_k/6$ and $C_0 - P\sigma_k/3$. The fleet LP may take $\chi \in [-1, 1]^N$ by Theorem 1(ii) and has $C^{\mathrm{LP}} = C_0$, attained at $\chi = 0$, since $\max\{\sigma_k/6, -\sigma_k/3\} \ge 0$. Hence

$$\begin{aligned}
C^\star - C^{\mathrm{LP}} &= \min_{\chi\in\{\pm1\}^N} \max_k \max \left\{ \frac{P}{6}\sigma_k, -\frac{P}{3}\sigma_k \right\} \\
&\in \left[\frac{P}{6}, \frac{P}{3} \right] \mathrm{disc}(T).
\end{aligned} \quad \text{(S7)}$$

For a Hadamard matrix H of order $N = 2^r$ with first row all ones and $T_k = \{i : H_{ki} = 1\}$, every $\chi \in \{\pm1\}^N$ with $\Sigma = \sum_i \chi_i$ has $\sigma_1 = \Sigma$, $(H\chi)_k = 2\sigma_k - \Sigma$ and $\|H\chi\|^2 = N^2$, so $\max_k |\sigma_k| \ge \max\{|\Sigma|, (\sqrt{N} - |\Sigma|)/2\} \ge \sqrt{N}/3$. By (S7), $C^\star - C^{\mathrm{LP}} \ge P\sqrt{N}/18 = \Omega(\sqrt{m}\, P_{\max})$. For set systems with $\Theta(N)$ sets on N elements, telling $\mathrm{disc}(T) = 0$ from $\mathrm{disc}(T) \ge c\sqrt{N}$ is NP-hard for some constant $c > 0$ [5]. By (S7), $C^\star = C^{\mathrm{LP}}$ in the first case and $C^\star \ge C^{\mathrm{LP}} + cP\sqrt{N}/6$ in the second, with $m = \Theta(N)$, so a polynomial algorithm with a peak guaranteed within $\epsilon\sqrt{m}\, P$ of $C^\star$, $\epsilon\sqrt{m} < c\sqrt{N}/6$, would separate them. □

V. Proof of Theorem 4

Lemma S2. *Let s_k be the k-th heat start. For all heat times within the bounds that agree with the observations up to slot t, the k-th tap lies in $[\underline{\eta}_t(s_k), \bar{\eta}_t(s_k)]$.*

Proof. The heats $k' \le k$ tap by $s_{k'} + \overline{d}_t(s_{k'}) \le \bar{\eta}_t(s_k)$, and the heats $k' \ge k$ tap at or after $s_{k'} + \underline{d}_t(s_{k'}) \ge \underline{\eta}_t(s_k)$, so at least k taps occur by $\bar{\eta}_t(s_k)$ and at most $k-1$ before $\underline{\eta}_t(s_k)$. □

1) Part (i): Let p be an integer point of $\mathcal{R}_t$, executed with round-robin units and first-in first-out ladles, and fix heat times within the bounds that agree with the observations up to t. With s_k and z_k the starts of p, rows (11a) at $s = s_k$, (11b) at $s = z_{k+q}$ and (11c) at $s = s_k$ give

$$s_{k+N^{\mathrm{E}}} \ge \bar{\eta}_t(s_k) + g, \qquad \bar{\eta}_t(s_k) \le z_{k+q} \le \underline{\eta}_t(s_k) + w.$$

As heat k taps by $s_k + \overline{d}_t(s_k) \le \bar{\eta}_t(s_k)$ and, by Lemma S2, the k-th tap lies in $[\underline{\eta}_t(s_k), \bar{\eta}_t(s_k)]$, these give (1d) and (1f), while (S1) covers the waiting ladles and (S2b) gives (1e). New observations give $\bar{\eta}_{t+1} \le \bar{\eta}_t$, $\underline{\eta}_{t+1} \ge \underline{\eta}_t$ and $\bar{\theta}_{t+1} \ge \bar{\theta}_t$, so, the counters being non-decreasing, every row of $\mathcal{R}_{t+1}$, including the fixed starts, holds at the point p_t executed at t. Hence every $\mathcal{R}_{i,t}$ is feasible, and by the argument above, the executed schedule, equal to p_T at the last slot T, is zero-break at the realized heat times, for every realization within the bounds.

2) Part (ii): By (S5), every combination of shifted floors that Algorithm 1 may execute lies in $\mathcal{R}_{i,t} \cap \mathbb{Z}$, so the algorithm respects $\mathcal{R}_{i,t}$ and part (i) applies. At an optimal vertex of (12), all left rows of (12b) are tight, as C_β has a positive cost in (12a), and if $\mu \le m_t$ right rows are tight, the face count of part (i) of Theorem 3 with the $1+m_t$ variables C and C_β gives $1 + m_t + \sum_i \dim G_i \le m_t + \mu$. At most $m_t - 1$ plants are thus fractional, the rest of that proof holds for the m_t remaining blocks with the peak of the vertex for C^{LP}, and the search accepts only lower peaks. □

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